

SCIENCE 10

Biology



Coverage: 3rd Quarter
Science 10 – Content Book

NAME: _____

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ABOUT THE MODULE



What this module is about?

UNIT III: LIVING THINGS AND THEIR ENVIRONMENT



Living organisms produce their own kind, transmit genetic information to their offspring, regulate and control their body processes, and respond to stimuli from the external environment. These abilities ensure their survival in a complex ecosystem.

Organisms have undergone a long process of evolution characterized by changes in their genetic material. Certain morphological and physiological characteristics have enabled them to adapt to different environments. In the housefly (*Musca domestica*) shown, for example, the presence of a compound eye is an adaptation for wider peripheral vision, while the modified spongelike tongue (labellum) is used to ingest liquefied food. The housefly's traits and other unique characteristics are the reasons they have survived 65 million years of existence, making them one of the most abundant and widely dispersed insects in the world.

In this unit, Chapter 10 introduces the complex reproductive organs of human males and females, and describes how sex cells are produced to ensure the perpetuation of the species. Chapter 11 looks at how the body regulates its internal processes to ensure homeostasis. Chapter 12 explores the details of the brain and spinal cord, and probes the processes by which the body controls its actions. It also examines how sense organs perceive stimuli and how they interact with the nervous system to elicit a coordinated response. Chapter 13 uncovers the mysteries behind inheritance and variation through a closer look at the chemical nature of genes. Chapter 14 describes how organisms adapt to changes in the environment in order to survive. Lastly, Chapter 15 examines the intricacies of the ecosystem and biodiversity, and how stability is maintained among populations of organisms.



Source:

- Silverio A. & De Castro-Bernas (2015), Ph.D; **The New Grade 10: Exploring Life through Science, 2nd Edition**, Phoenix Publishing House, Inc.



LESSON PROPER:

Lesson 10.4

Major Endocrine Glands in the Body

Main Idea: Each endocrine gland releases hormones that control specific processes.

EQ: What important roles do endocrine glands perform in the body?

The Hypothalamus

The hypothalamus is found at the base of the brain. It produces hormones that help turn the other endocrine glands in your body on and off. It is the link between the nervous system and the endocrine system. Therefore, the hypothalamus knows about the sensations you are aware of, such as the smell of a beautiful flower or a picturesque view. At the same time, it also controls those things that you are not aware of, such as the level of hormones in your body or your internal temperature. The hypothalamus secretes hormones that stimulate or suppress the release of the hormones in the pituitary gland.

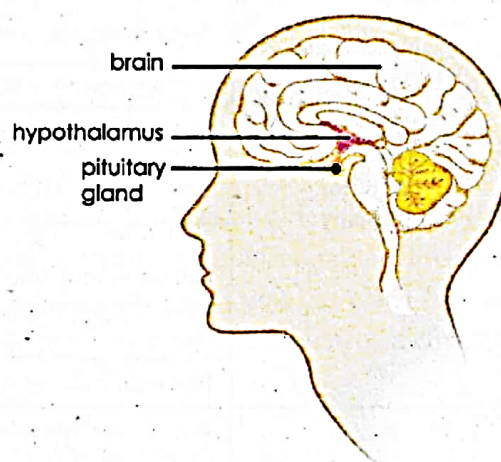


Fig. 10-6. Hypothalamus and pituitary gland

The Pituitary Gland

The pituitary gland or hypophysis is a pea-sized gland found in the center of the skull, right behind the bridge of the nose and below the hypothalamus of the brain. The hypothalamus communicates with the pituitary gland through nerve impulses and hormones. The hypothalamus is responsible for triggering the production of hormones by the pituitary gland. For many years, the pituitary was referred to as the "master gland" because it produces hormones that control many activities of the other endocrine glands. With the discovery that the pituitary was controlled by the hypothalamus, the nickname "master gland" gradually dropped out of use.

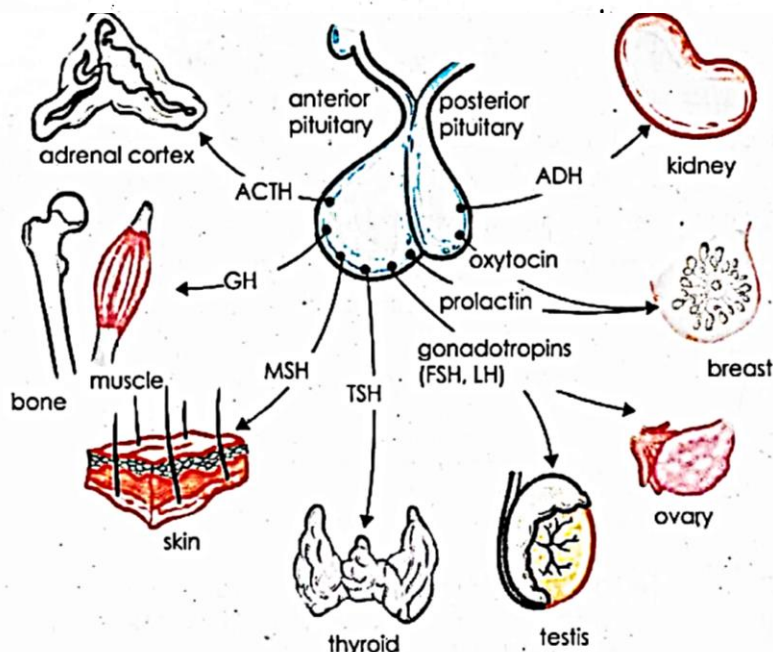


Fig. 10-7. Anterior and posterior pituitary glands and their secretions

The pituitary gland plays a very important role in the endocrine system since it releases many hormones that control other endocrine glands. It is made of two lobes, the anterior and posterior glands, each secreting different hormones.

Figure 10-7 shows the specific hormones secreted by the anterior and posterior pituitary glands. The pituitary also secretes *endorphins*, the hormones responsible for reducing feelings of pain.

Table 10-1 Hormones Secreted by the Pituitary Gland

	Pituitary Hormones	Action
Anterior pituitary	Adrenocorticotrophic hormone (ACTH)	Stimulates the adrenal glands to produce related steroid hormones
	Growth hormone (GH)	Stimulates the growth of bones and tissues
	Melanocyte-stimulating hormone (MSH)	Regulates melanin production
	Thyroid-stimulating hormone (TSH)	Stimulates the thyroid gland to produce thyroid hormone
	Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)	Controls sexual function and production of the sex hormones: estrogen and progesterone in females, or testosterone in males
	Prolactin	Stimulates milk production in women
Posterior pituitary	Antidiuretic hormone (ADH) or vasopressin	Controls water loss in the kidneys
	Oxytocin	Stimulates milk ejection or secretion and contraction of the uterus during childbirth

The Thyroid Gland

The thyroid gland is located just under the larynx or voice box. It secretes three hormones: triiodothyronine (T_3), thyroxine (T_4), and calcitonin. Thyroid hormones also help in maintaining heart rate, blood pressure, muscle tone, and reproductive functions. Both T_3 and T_4 contain the element iodine, which regulates metabolism (how quickly food is burned up in the body) and plays a crucial role in development and maturation. An excess of T_3 and T_4 in the blood can lead to a condition known as *hyperthyroidism*, which can make a person develop high blood pressure, feel irritable, sweat profusely, and lose weight. An insufficient amount of these hormones in the body can lead to *hypothyroidism*, which causes weight gain, lethargy, and intolerance to cold. Severe iodine deficiency in the diet can cause goiter in adults. Calcitonin, on the other hand, lowers the blood calcium level in the body. Calcium is a mineral that keeps the nerves and muscles working properly.

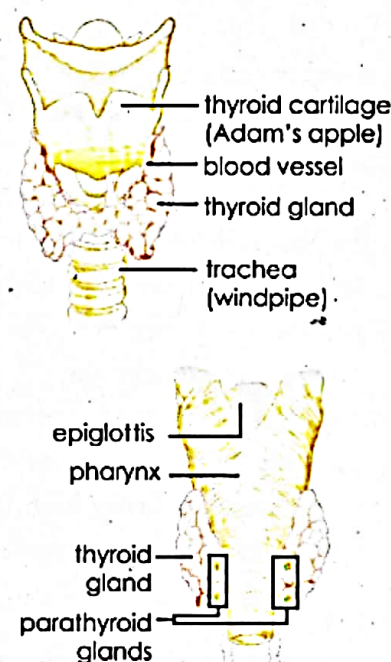


Fig. 10-8. Thyroid and parathyroid glands

The Parathyroid Glands

The parathyroid glands are four tiny glands embedded in the surface of the thyroid. These glands release the *parathyroid hormone* which, along with calcitonin, regulates blood calcium levels in the body in an antagonistic manner since they have opposite effects.



Scan the QR code to watch a suggested video on the location and function of the thyroid and parathyroid glands.

Thymus

The thymus is responsible for the development of the defense system of the body against disease-causing organisms. It is found just behind the breastbone. The thymus produces and develops T cells, a type of white blood cells that protect body tissues by triggering an immune response against infectious agents. At a young age, your thymus is large but it gradually decreases in size as you get older because other glands in the body take over its job. By adulthood, the thymus will have already shrunk to about the size of a thumb.

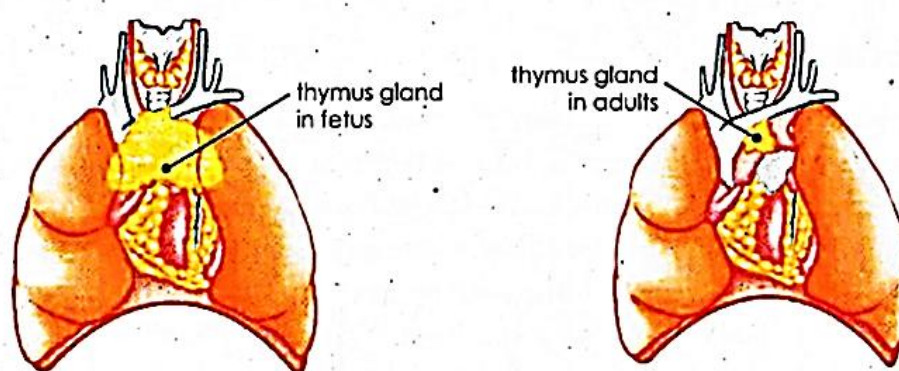


Fig. 10-9. Thymus glands of a fetus and an adult

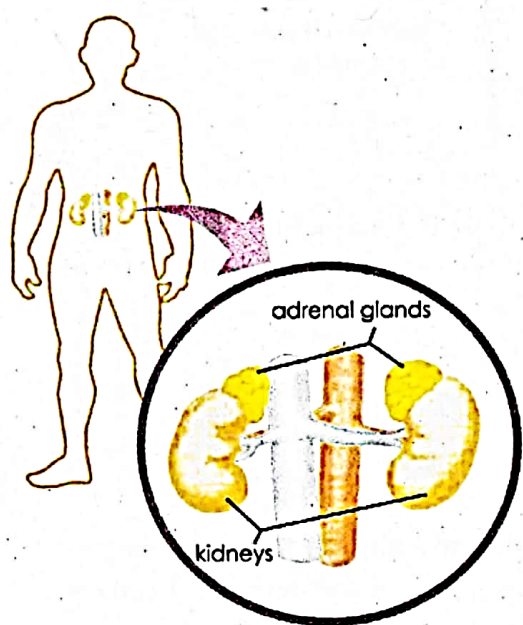


Fig. 10-10. Adrenal glands

Adrenals

The word *adrenal* means "above kidney." Each adrenal gland is located atop each of your kidneys. The adrenal glands are made up of two parts: the inner adrenal medulla and the outer adrenal cortex. The inner adrenal medulla produces *adrenaline* which plays an important role in the body's coping mechanism against short-term physical and emotional stress. It is part of the body's emergency action team. Whenever you are in a dangerous situation, the senses send signals to the brain that alerts the body and warns you of danger. The brain shocks the body into action to avoid the threat. It also alerts the adrenal glands to produce adrenaline, which at times gives the body a strength that could be considered "superhuman." For example, when there is a fire outbreak, some people can move or lift heavy objects that they cannot under normal circumstances. The outer adrenal cortex secretes corticosteroids which regulate the body's metabolism, electrolyte balance, immune system, and long-term stress response.

The Pancreas

Although part of the digestive system, the pancreas is also an endocrine organ. It produces two types of hormones: insulin and glucagon. Insulin is produced by a small group of cells called *islet of Langerhans* within the pancreas. Insulin keeps the levels of sugar (glucose) in the bloodstream under control. It does this by helping cells absorb the sugar and use it for energy. It also helps change excess sugar into a substance called glycogen, which can be stored in the liver for later use. This loop prevents the level of sugar in the body from rising too high. Without enough insulin, a person can develop *diabetes mellitus*, a disorder in which the level of sugar in the blood is too high. Glucagon, on the other hand, does exactly the opposite. It increases the level of sugar in the body by speeding up the conversion

of glycogen into sugar in the liver. Both these hormones ensure that the amount of sugar in the blood is always the right level. If the sugar level drops, the pancreas releases more glucagon to make up for the loss. If the sugar level rises, the pancreas releases more insulin to get rid of excess sugar.

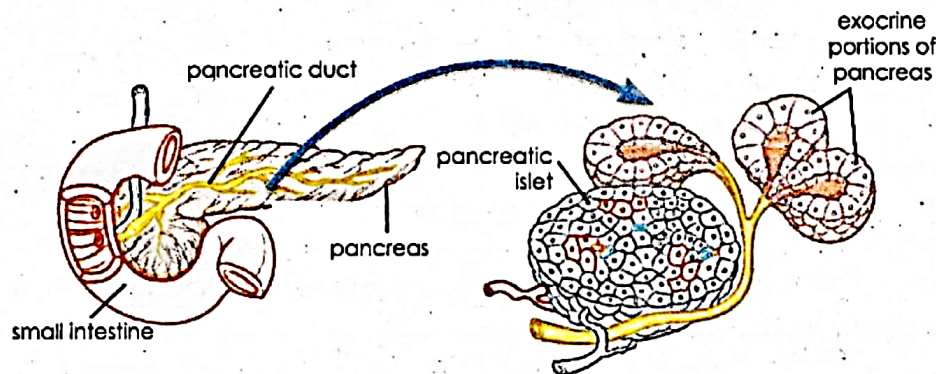


Fig. 10-11. Pancreas and close-up view of islets of Langerhans that produce and secrete insulin and glucagon hormones

The Gonads

The human reproductive organs are collectively called gonads. The reproductive organs in females are the *ovaries*, while testes in males. Both the testes and ovaries produce sex hormones responsible for the development of secondary sexual characteristics during teenage years. The testes produce androgens, of which the most important is testosterone. These male hormones induce the development of facial hair, body mass, and deep voice, and the production of sperm. The ovaries produce estrogen and progesterone that control the development of breasts, menstruation, and fertility. Other female reproductive structures such as the placenta and uterus release other hormones during pregnancy and childbirth.



Sci Bit

Osteoporosis is a bone disorder commonly experienced by aging people. It makes the bones less dense and prone to fractures. Did you know that this condition is attributed to an underlying endocrine problem? Postmenopausal women develop this disease due to the diminishing production of the hormone estrogen, which helps in maintaining bone mass. Hormone replacement therapy is usually prescribed for aging women to reduce the risk of osteoporosis.

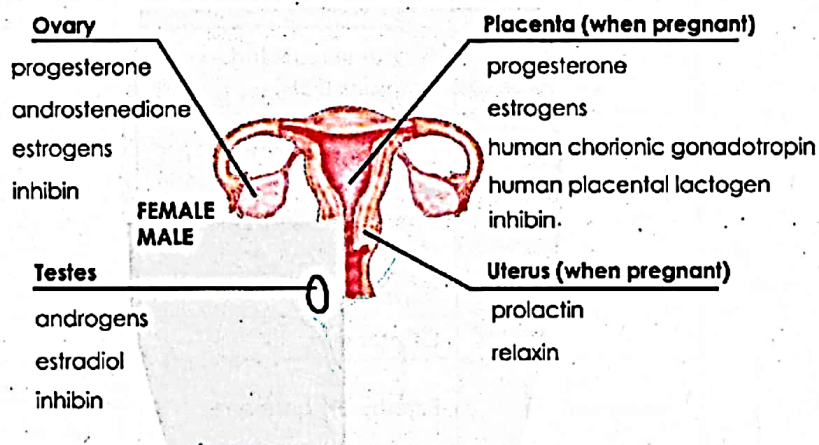


Fig. 10-12. Gonads and their hormones

ADDITIONAL NOTES:

The Pineal Gland

The pineal gland is located near the thalamus of the brain, where the two halves of the brain join. It produces *melatonin*, which regulates body rhythms and sleep patterns. It synchronizes daily rhythms, regulates body systems, and helps the body adapt to seasonal changes. Jet lag, the condition resulting from traveling across multiple time zones, disrupts sleep or activity patterns due to the inability of the body to immediately reset its internal circadian clock to external environmental rhythm. This desynchronized sleep-wake cycle is corrected with the amount of melatonin produced during daytime and nighttime.

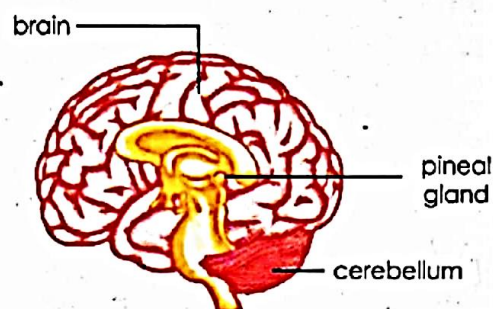


Fig. 10-13. Pineal gland



Scan the QR code to watch a suggested video and learn more about the pineal gland.

Table 10-2 Major Endocrine Organs and Hormones

Tissue/Gland	Hormone Produced	Target	Major Action
Hypothalamus	Releasing and inhibiting hormones	Anterior pituitary	Stimulates or inhibits the release of specific pituitary hormones
Anterior pituitary	Thyroid-stimulating hormone (TSH)	Thyroid	Stimulates production and secretion of thyroxine
	Prolactin	Mammary gland	Stimulates milk production
	Adrenocorticotrophic hormone (ACTH)	Adrenal cortex	Stimulates secretion of corticosteroids
	Endorphin	Brain	Decreases pain
	Growth hormone (GH)	Various cells	Stimulates general body growth
	Luteinizing hormone (LH)	Ovary	Stimulates ovulation and production of estrogen and progesterone
		Testes	Stimulates sperm production
Posterior pituitary	Oxytocin	Mammary gland	Stimulates milk secretion
		Uterus	Induces uterine contraction
	Antidiuretic hormone (ADH)	Kidney	Increases water absorption
Thyroid	Thyroxine	Most cells	Increases metabolic rate and growth
	Calcitonin	Bones	Stimulates calcium uptake
Parathyroid	Parathyroid hormone	Bones	Stimulates calcium release into blood
		Digestive tract	Stimulates calcium absorption

Adrenal glands	Epinephrine (adrenaline) and norepinephrine (noradrenaline)	Circulatory system	Increases heart rate, blood pressure, and blood sugar
		Respiratory system	Increases breathing rate and clears airways
Pancreas	Insulin	Various cells	Stimulates glucose uptake from blood
	Glucagon	Various cells	Stimulates glucose release from cells into blood
Ovary	Estrogen	Various cells	Stimulates female development and behavior
	Progesterone	Uterus	Stimulates growth of uterine lining
Testes	Testosterone	Various cells	Stimulates male development and behavior
Thymus	Thymosin	White blood cells	Stimulates differentiation in white blood cells
Pineal	Melatonin	Brain	Promotes sleep
Gastrointestinal tract	Gastrin	Gut cells	Stimulates hydrochloric acid secretion

Lesson 11.4

Divisions of the Nervous System

Main Idea: The central and peripheral nervous systems work together to help the body respond to stimuli.

EQ: What makes you react appropriately to everything around you?

The previous section identifies the neuron as the basic unit of structure and function of the nervous system. It also discusses some insights about the amazing job neurons do to keep the body in sync with the events happening in the body's external and internal environments. Neurons do not act in isolation; they work with other neurons to form a complex communication network that makes up the human nervous system.

Open The nervous system is divided into two main parts: the *central nervous system* and *peripheral nervous system*. Figure 11-6 illustrates the two divisions of the nervous system working together as a team to monitor, coordinate, and control the activities of the entire body.

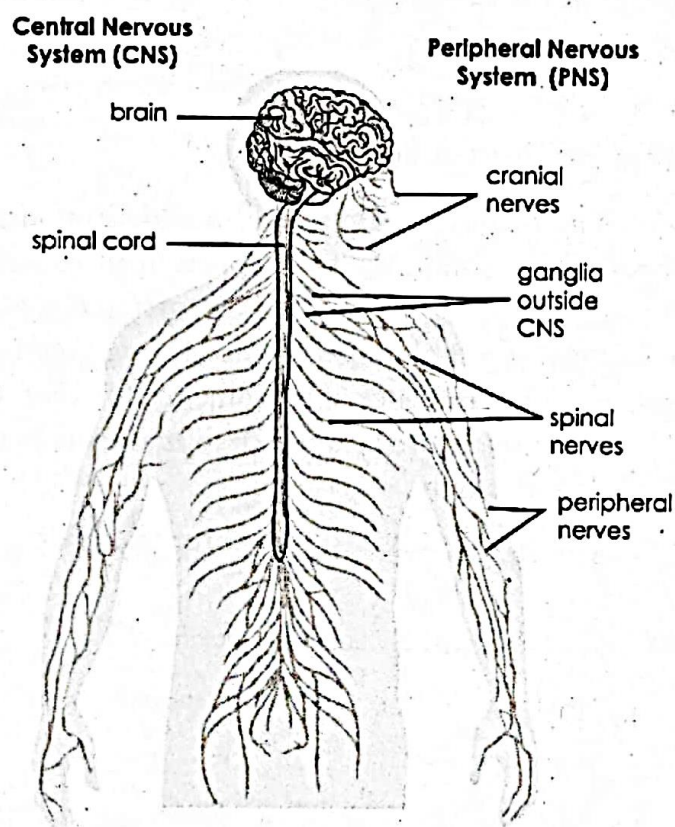


Fig. 11-6. The divisions of the nervous system in the human body

The Central Nervous System

The central nervous system (CNS) is the command center of the entire body; it processes information and sends instructions to other parts of the body. All the information gathered by sensory receptors is delivered to the central nervous system in the form of nerve impulses. The CNS processes the information and responds by sending nerve impulses to the motor nerves of the peripheral nervous system (PNS). Interpreting the information and issuing appropriate commands are the responsibility of the two parts of the central nervous system: the brain and the spinal cord. The *brain* is the main control center of the central nervous system. It transmits and receives messages through the spinal cord. The *spinal cord* provides the link between the brain and the rest of the body.

The Brain

The brain is an important and delicate organ that is protected by and encased in a bone called the *skull*. The brain is also wrapped in three layers of connective tissue that nourishes and protects it. It is also bathed with fluid called *cerebrospinal fluid* (CSF) that cushions it against sudden impact such as bumps or tumbles. The brain appears gray because it is mostly composed of gray matter, which is because of the presence of billion of cell bodies. Underneath this gray matter is white matter, which is due to the myelinated axons. Despite the presence of billions of neurons, the mass of the brain is about 1.4 kilograms only.

The brain is divided into three main parts: the cerebrum, cerebellum, and brain stem.

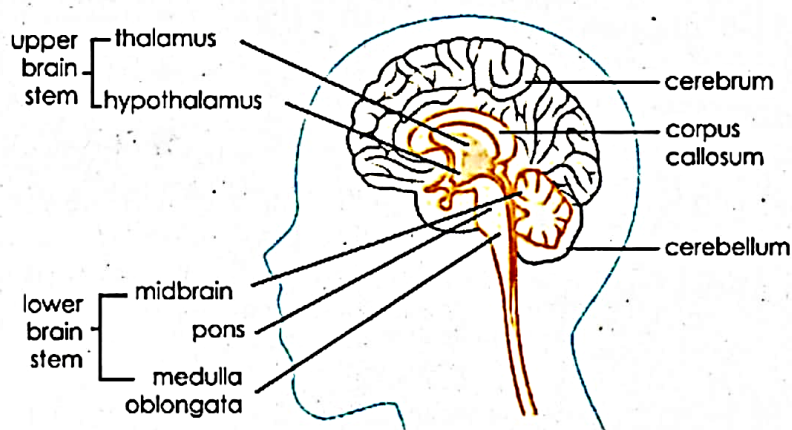


Fig. 11-7. The human brain. This view shows the left hemisphere of the brain.

The Cerebrum

The cerebrum makes up 85 percent of the human brain, making it the most noticeable part of the organ. It is also what most people often refer to when they see a picture of the brain.

The surface of the cerebrum is called the *cerebral cortex*, which is lined with deep, wrinkled grooves that increase the surface area so that more thinking activities can occur. This is the area where learning, intelligence, and judgment occur. Aside from this enormous task, the cerebral cortex also controls all the voluntary (under one's control) activities of the body. In addition, it shapes the attitudes, emotions, and even the personality of an individual.

As shown in figure 11-8, the cerebrum is divided into two halves. Each hemisphere controls different kinds of mental activity. For example, the right hemisphere is associated with artistic ability, and the left hemisphere with mathematical or logical ability. Note that each half controls the movement of, and sends sensations to, the opposite side. In other words, the right side of the brain controls the left side of the body, while the left side of the brain controls the right side of the body.

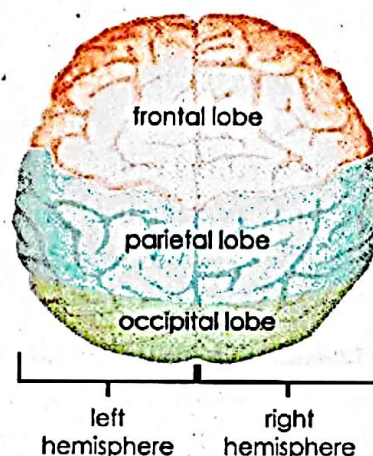


Fig. 11-8. The top view of the brain, showing the cerebral hemispheres

The Cerebellum

The cerebellum is the second largest part of the brain and is located behind the brain stem. It coordinates the actions of the muscles and maintains balance so that the motor movements of the body can move smoothly, steadily, and efficiently. Balance, equilibrium, and posture are coordinated by the cerebellum.

The Brain Stem

The *brain stem* connects the spinal cord to the rest of the brain. It coordinates many survival functions of the body such as breathing, heart rate, sleep, and wakefulness. Three distinct regions make up the brain stem: the medulla oblongata, the midbrain, and the pons. The *medulla oblongata* controls involuntary actions such as heartbeat, breathing, and blood pressure. The *midbrain* processes visual and auditory reflexes. The *pons* helps control respiratory functions. Two other important structures are found in the brain stem: the thalamus and the hypothalamus. The *thalamus* serves as a relay station by directing incoming messages from the spinal cord to the appropriate parts of the brain. The *hypothalamus*, on the other hand, monitors internal conditions such as water content and temperature. It acts as the link between the endocrine and nervous systems.



What are the three main parts of the brain?

The Spinal Cord

The spinal cord is a tubelike organ of neurons and blood vessels. It is located inside the backbone and is about 1.8 cm wide, nearly the same size as a garden hose. Aside from the bones of the spine, the spinal cord is also covered with protective membranes called *meninges*. The spinal cord relays nerve impulses to and from the brain. Impulses from the peripheral nervous system are carried to the spinal cord, which then brings the message to the brain. The brain, in turn, sends impulses down the spinal cord to the motor nerves of the peripheral nervous system.

As an example, you might not notice a mosquito hovering around your face when you are so engrossed in reading. But if the mosquito happens to come close to your eyes, you instantly blink (reflex) in response to a stimulus (mosquito coming near your eyes). In this situation, the reflex kicks in as soon as the mosquito approaches your eyes. The mosquito's action sends a nerve impulse via sensory neurons to the spinal cord which is relayed to interneurons, and then to the motor neurons. The motor neurons stimulate the muscles of the eyes, causing them to contract and make the blinking motion.

Reflexes are not only lightning-fast reactions; they are also automatic. The speed and nature of a reflex is possible because the nerve impulses bypass the brain and travel directly to the spinal cord. The brain becomes aware of the event only after it has happened.

A similar sequence happens when you touch a hot object. Your automatic response is to quickly withdraw your hand from a hot object. The neurons, in a pathway called the *reflex arc*, carry the nerve impulses for an automatic response. Figure 11-9 shows the reflex arc when you remove your hand from a hot object. This arc consists of several events.

1. The heat on the object stimulates a heat receptor in the skin of the hand.
2. Impulses that are produced travel from a sensory neuron to the spinal cord.
3. In the spinal cord, the impulses are transmitted first across a synapse through an interneuron, which then carries a nerve impulse to a motor neuron.
4. The motor neuron then carries the nerve impulse to a muscle.
5. The muscle contracts and pulls the hand away from the hot object.

Again, notice that the interneuron in this pathway is in the spinal cord, not in the brain. A reflex that does not involve the brain is called a *spinal reflex*. Spinal reflexes enable you to respond quickly because you do not have to think about it.

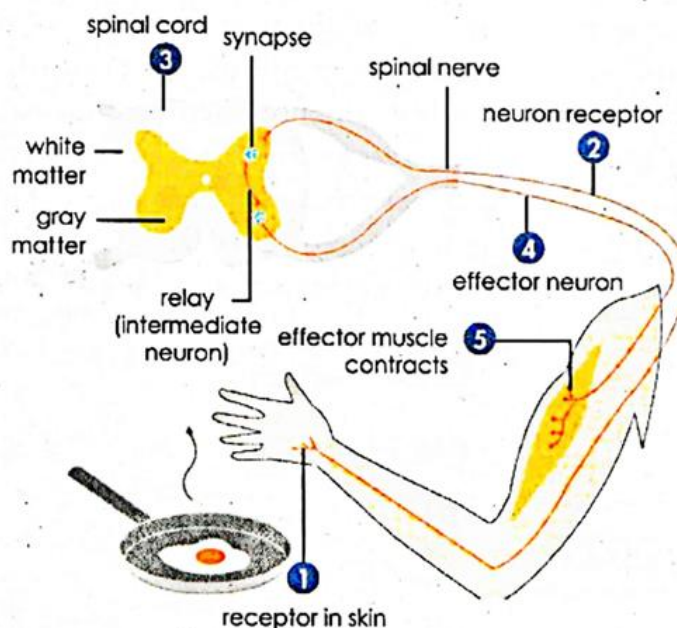


Fig. 11-9. Reflex arc



Scan the QR code to watch a suggested animation on how a reflex happens.

The Peripheral Nervous System

The peripheral nervous system (PNS) links the central nervous system (brain and spinal cord) to the rest of the body. It is made up of a network of bundled ropelike structures called *nerves*, which extend or branch out from the central nervous system to the different organs of the body. The PNS gathers and delivers information to and from the central nervous system.

The peripheral nervous system consists of 43 pairs of nerves that arise from the brain and spinal cord, leading to organs throughout the body. Many of the nerves in the PNS are under the direct control of the conscious mind. For example, when you want your leg to move, a message from your brain travels to your spinal cord, goes through a peripheral nerve, and then to your leg. The part of the peripheral nervous system that stimulates skeletal muscles under your conscious control is called the *somatic nervous system*. The somatic nervous system is also responsible for the mediation of spinal reflexes, which are involuntary. Spinal reflexes are extremely rapid; they involve the spinal cord and the PNS but not the brain. The part of the peripheral nervous system that is not under the direct control of the conscious mind is called *autonomic nervous system*. The autonomic nervous system controls body activities that are involuntary—that is, those that happen automatically without thinking about them. The autonomic nervous system directs motor nerve fibers in smooth muscles, cardiac muscles, and the glands. Contractions of the heart muscle and movement of smooth muscles surrounding the small intestine are activities under the control of the autonomic nervous system.

The nerves of the autonomic nervous system have two divisions that have opposite effects on the organs they control. Their antagonistic nature maintains stability in the body by keeping the activities in balance.

The *parasympathetic division* is most active under normal conditions; it keeps the body functioning even when the person is not active. For instance, your breathing continues even while you are sleeping. The sympathetic division dominates in times of physical and emotional stress or whenever the body requires some action. It controls the “fight-or-flight” response during stressful situations such as the nervousness you may experience when speaking before an audience or taking a surprise quiz. When you are frightened, the nerves leading to organs, such as the lungs and the heart, are activated. The *sympathetic division* increases blood pressure, heart rate, and breathing rate. Such increase may be necessary if extra energy and strength are needed to deal with the stressful situation. But when the situation is over, the parasympathetic nerves bring the blood pressure, breathing rate, and heart rate back to normal:

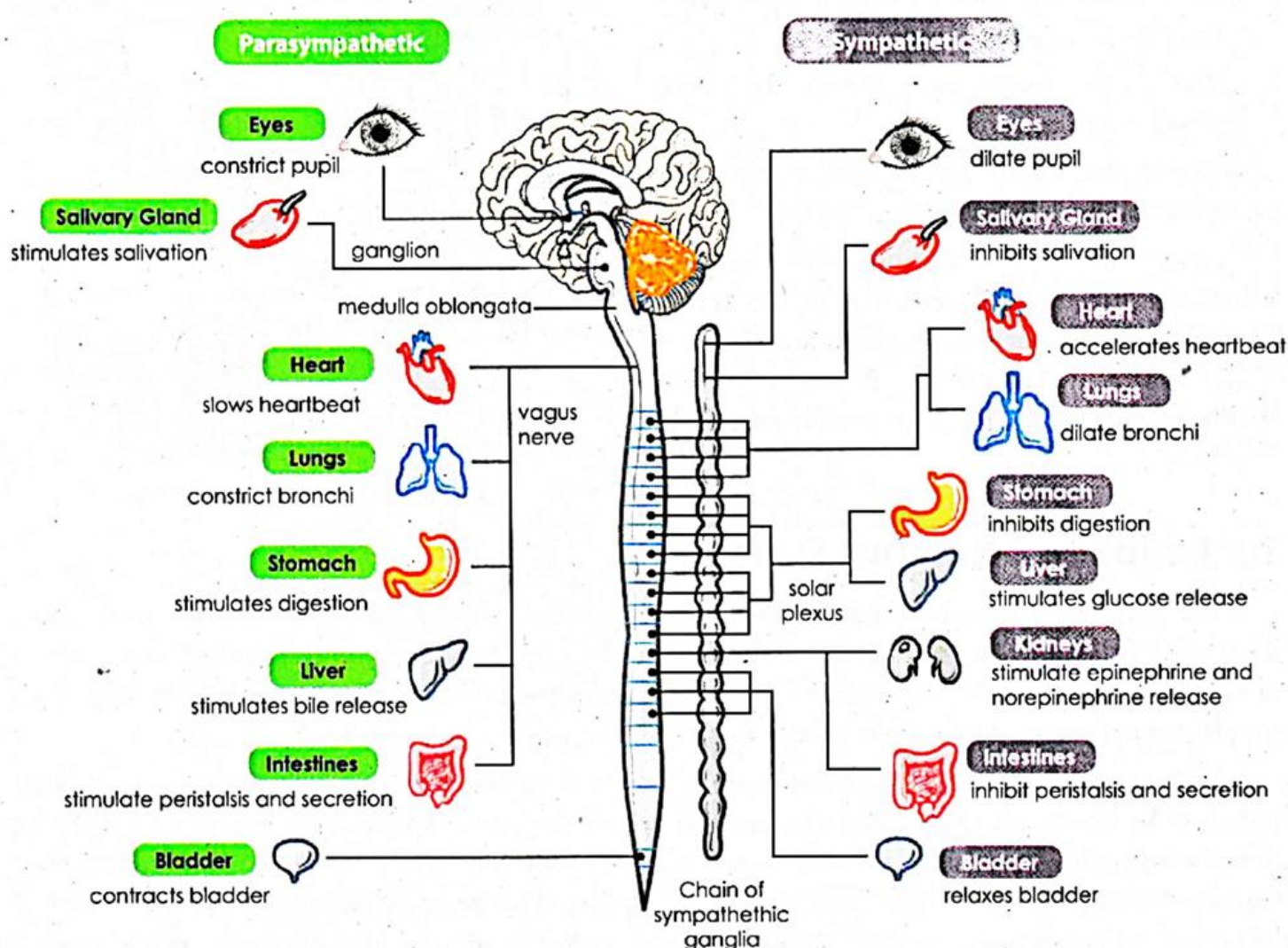
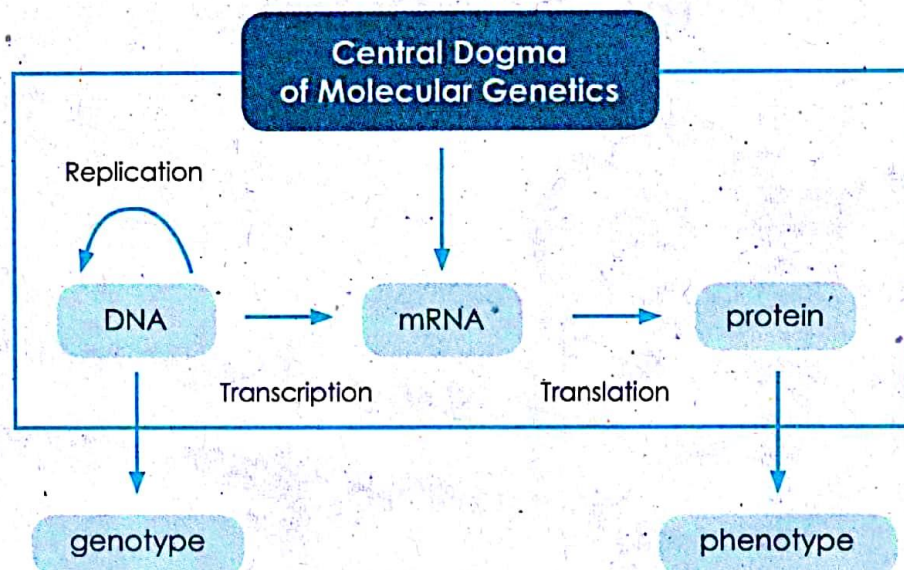


Fig. 11-10. Physiological effects of the autonomic nervous system

ADDITIONAL NOTES:



Concepts in a Box



Lesson 12.1

DNA Structure

Main Idea: Experimental studies revealed that DNA is the molecule of heredity.

EQ: Why is understanding the structure of DNA important?

The experiments of Gregor Mendel established the basic principles of heredity. These principles gave rise to more questions that led to several significant researches. Upon the turn of the twentieth century, with the advent of biochemistry and molecular biology, the chemical nature of the gene was slowly elucidated. Today, it is known that the genetic material is found in chromosomes.

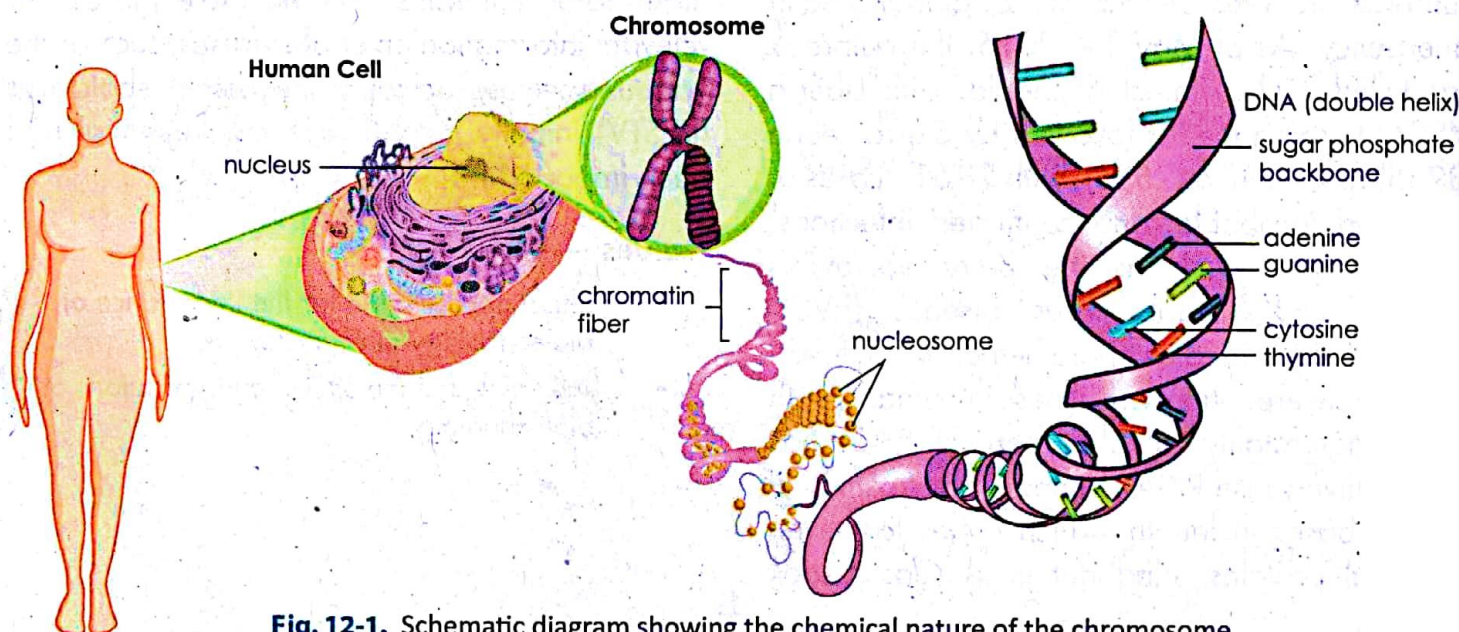


Fig. 12-1. Schematic diagram showing the chemical nature of the chromosome

The Watson-Crick DNA Model: A Double Helix

On April 25, 1953, an article in the scientific journal *Nature* titled "Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid" was published. The two-page article was authored by American biologist James Watson and English physicist Francis Crick. This was the first published article that described the structure of the DNA as a double helix. Considered as a "pearl" of science, the article contained the answers to how the genetic information inside the nucleus of cells was stored and passed from one generation to the next. The article was considered a giant scientific leap and a turning point in the development and rapid progress of molecular biology and genetics.

The structure of DNA proposed by Watson and Crick was the amalgamation of numerous findings from various scientists. The model was based on the x-ray diffraction image taken by Rosalind Franklin and Raymond Gosling in 1952. The pairing of the DNA bases, where adenine pairs with thymine and cytosine pairs with guanine, were based on Chargaff experiments. The presence of *in vivo* (inside the living body) double-helix DNA structure was reported by Maurice Wilkins in the same issue of *Nature* in 1953. In recognition of the tremendous impact of the elucidation of the double-helix structure of DNA, James Watson, Francis Crick, and Maurice Wilkins were awarded the Nobel Prize in Physiology or Medicine in 1962.

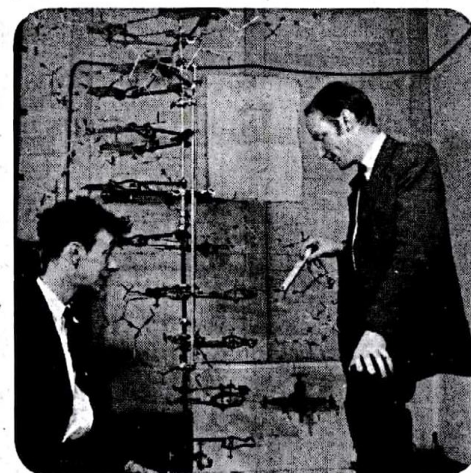


Fig. 12-6. Francis Crick (left) and James Watson (right) with their model of the famous double-stranded helical DNA structure

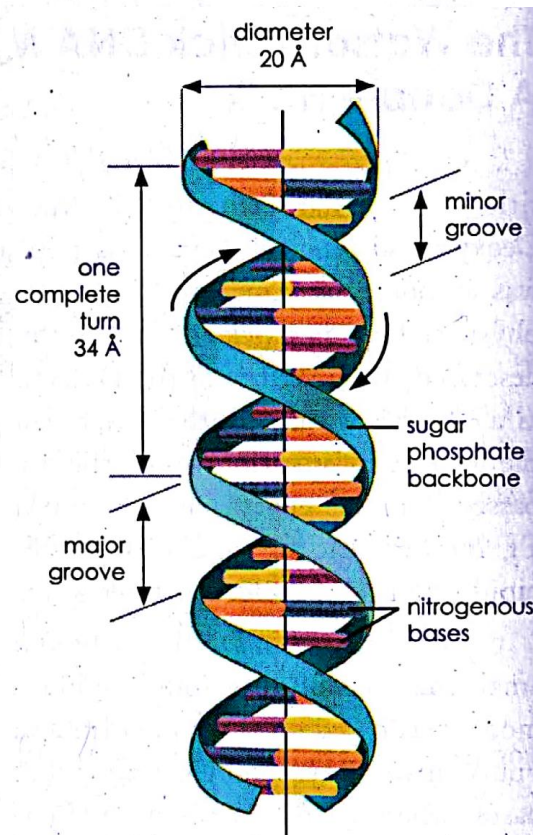


Fig. 12-7. The double-helical structure of DNA showing the positions of the sugar, the phosphate group, and the nitrogenous bases as proposed by James Watson and Francis Crick



Scan the QR code to view a suggested image of the original DNA model of Watson and Crick in 1953.

The structure of the DNA molecule as described by Watson and Crick showed that the two strands that make up a DNA molecule are wound around each other, forming a double spiral molecule resembling a twisted ladder. The backbone of the helix consists of alternating sugars and phosphates, while the steps of the ladder are made up of nitrogenous base pairs. According to the Watson-Crick model, the different nitrogenous bases form specific pairs, such that only A pairs with T, while C pairs with G. As a result of this specific pairing, the double-stranded DNA molecule forms a uniform structure within the entire length of a long helix. Hydrogen bonds connect the nitrogenous base pairs together, making the double helix highly stable. There are three hydrogen bonds between C and G, and two hydrogen bonds between A and T.

Lesson 12.2

The Nucleic Acids and Their Connection with Inheritance

Main Idea: The nucleic acids are responsible for carrying specific instructions from the genes.

EQ: Why are nucleic acids essential in heredity?

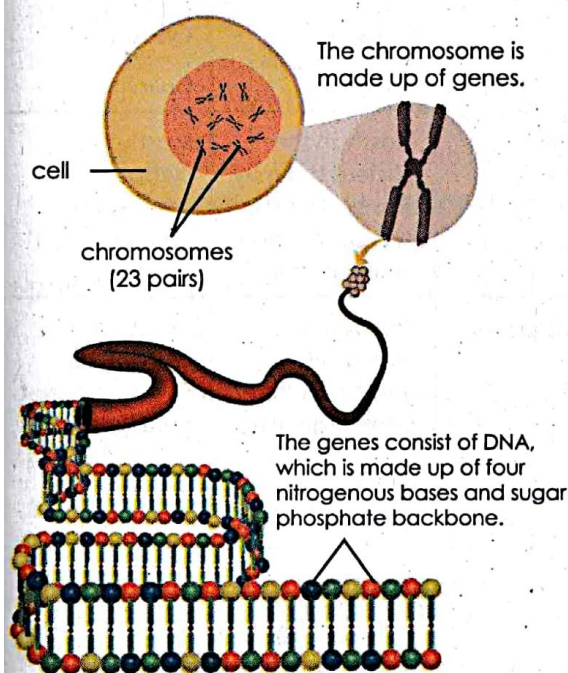


Fig. 12-8. The long strand of DNA is wound within the chromosome inside the nucleus of the cell.

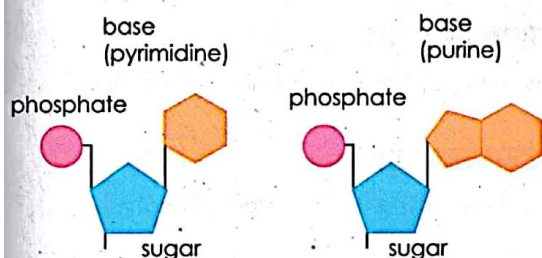


Fig. 12-9. Schematic representations of two nucleotides, one with a pyrimidine base and the other with a purine

Have you ever wondered why you are so complex? Your body is able to function in growth, cell repair, or development even without your conscious effort. How does the body know when to perform such bodily activities? The answer lies in specific sets of instructions inside your DNA.

Every human body cell contains 23 pairs of chromosomes or a total of 46 chromosomes. A single chromosome contains many genes joined together like beads on a string. The genes are packed in bundles of these chromosomes. Body cells contain about 20 000 to 25 000 genes. A *gene*, as shown in figure 12-8, is a distinct portion of the DNA responsible for an inherited trait. Genes are coded instructions for everything that must happen in the body, including how an individual functions and how a person looks. Normally, one gene controls one trait, but some traits are coded by more than one gene.

Nucleic acids are organic compounds that function as storage of genetic information, which is transmitted from one generation to the next in all living organisms. It is the physical carrier of inheritance that is passed from parents to offspring. Nucleic acids also function in protein synthesis as they carry the code needed in the formation of specific proteins. There are two types of nucleic acids found in living organisms: *deoxyribonucleic acid* (DNA) and *ribonucleic acid* (RNA). Both types are made up of basic building blocks called *nucleotides*. A nucleotide is made up of a five-carbon sugar, a phosphate group, and a nitrogenous base. The nitrogenous bases are either double-ringed purines (guanine and adenine) or single-ringed pyrimidines (cytosine, thymine, and uracil).

ADDITIONAL NOTES:

James Watson and Francis Crick described the structure of DNA as a double helix of repeating nucleotides that are made up of a sugar (deoxyribose), a nitrogenous base that is either a purine (adenine and guanine) or a pyrimidine (thymine and cytosine), and a phosphate group. The pairing of nitrogenous bases is so specific that only adenine pairs with thymine, while only cytosine pairs with guanine. Thus, a gene refers to a specific sequence of nitrogenous bases that codes for a specific protein. DNA can be compared to a blueprint of guidelines that the body must follow to exist and function properly. RNA, on the other hand, helps to carry out the blueprint's guidelines. RNA is able to perform a variety of functions and thus, is more diverse. DNA is able to carry complex information for longer periods of time and is thus more stable.

☒ self-check | How are DNA and RNA alike?
How are they different?

Table 12-1 Differences Between DNA and RNA

	DNA	RNA
Sugar	Deoxyribose (C ₅ H ₁₀ O ₄)	Ribose (C ₅ H ₁₀ O ₅)
Strand	Double-stranded	Single-stranded
Nitrogenous bases	Adenine Thymine Cytosine Guanine	Adenine Uracil Cytosine Guanine
Location	Mostly in the nucleus but may also be found in cytoplasm and mitochondria	Mostly in the cytoplasm but may also be found in the nucleus
Function	Blueprint of biological guidelines that living organisms must follow to exist and function properly	Assists in carrying out DNA's blueprint guidelines

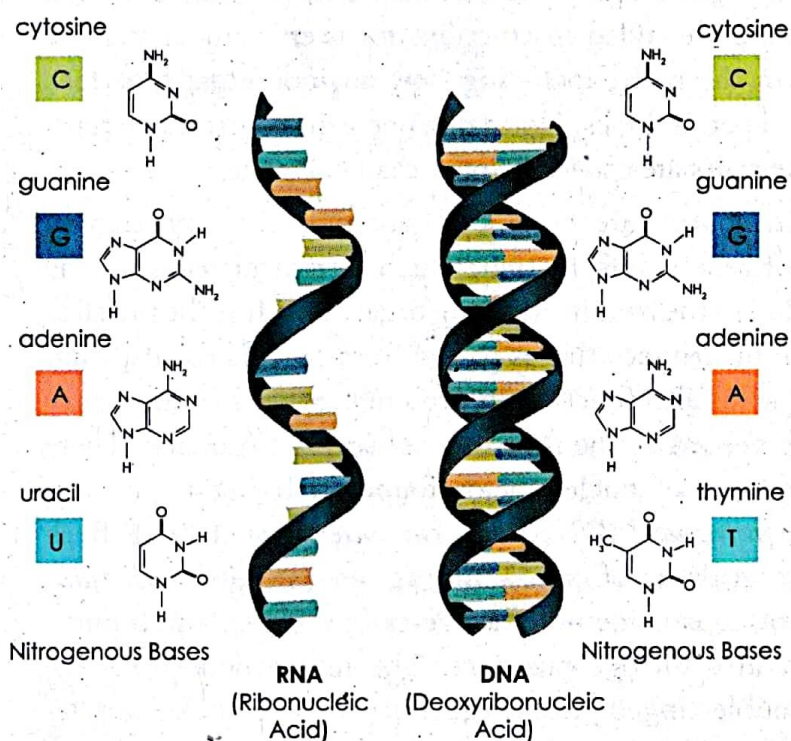
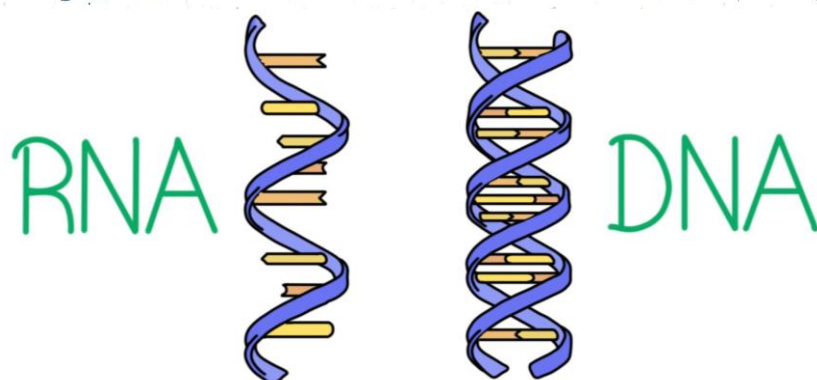


Fig. 12-10. Illustration of RNA and DNA structures



DNA is STORED.

Organized into genes and packaged in chromosomes

DNA is REPLICATED.

Duplicates genetic information with high fidelity

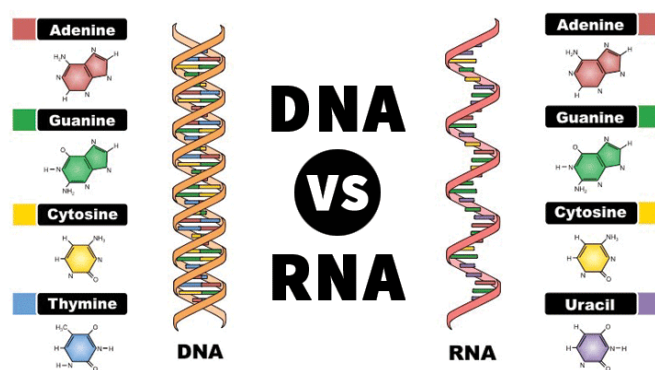
DNA is EXPRESSED.

Provided with a mechanism to affect phenotypes

DNA is DIVERSIFIED.

Ability to mutate to produce variation and drive evolution

Fig. 12-11. Key features of DNA as genetic material



Lesson 12.3

The Central Dogma of Molecular Genetics

Main Idea: The sequence of bases in a gene forms a code that informs the cell what protein to produce.

EQ: Why must a cell produce proteins?

Recall that a gene is a portion of the DNA. Specifically, a gene's function is to control the production of proteins inside the cells of organisms. As seen in figure 12-12, a gene is made up of a series of bases arranged in a specific order. In reality, a single gene may contain about several hundreds to millions of nitrogenous bases. The order of the nitrogenous bases in a specific gene is the *genetic code* that gives instructions on what protein will be produced.

In genetics language, the material found inside the nucleus which makes up an organism's complete set of genes called *genotype* must be expressed as an observable characteristic or *phenotype*. This means proteins (phenotype) are produced using the language coded in the DNA (genotype). The process is summarized in the "central dogma of molecular genetics." The flow of genetic information is from DNA to messenger RNA (mRNA) to protein.

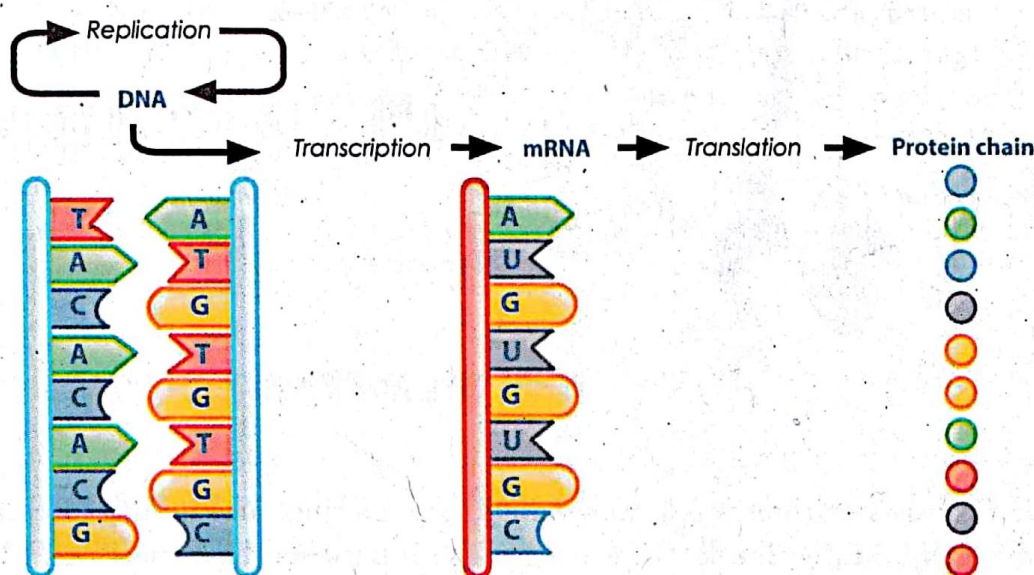


Fig. 12-12. Central Dogma of Molecular Genetics: Hereditary instructions (genotypes) are transcribed from DNA to RNA, then translated into proteins responsible for the formation of a phenotype of the organism.

The series of processes involved in the production of proteins is called *protein synthesis*. It involves the processes of replication, transcription, and translation.

Replication: How DNA Copies Itself

Cells need to make identical copies of their genetic material for growth and repair. Replication happens before cell division because each new cell is required to have an exact copy of the parent cell's DNA. Without replication, the new cell may not function properly. The complementarity of the nitrogenous bases makes it possible for the DNA to copy itself. One way to imagine the process of replication is to examine a cell prior to cell division. During the G1 stage of interphase, every chromosome in a cell is composed of one chromatid. Only during the next phase of interphase, the S phase, is the sister chromatid synthesized, resulting in chromosomes composed of two chromatids or sister chromatids. The S phase, or synthesis stage, is the process of replication.

During replication, the two DNA strands connected by hydrogen bonds separate from each other. Each old strand of the parent DNA is then used as a template for the building of a new strand in the daughter DNA. This process is called a *semiconservative replication* because one of the complementing strands is conserved in either daughter DNA.

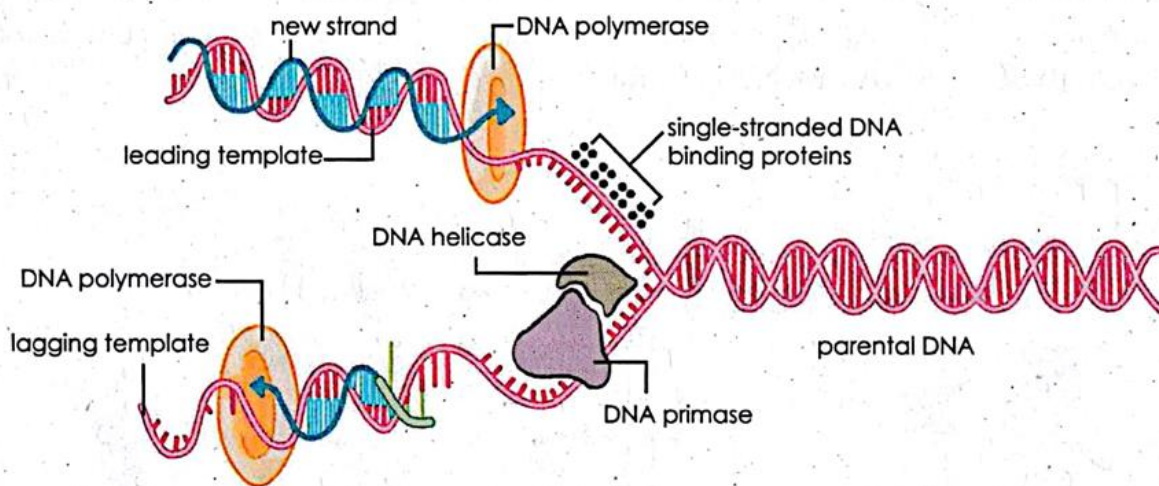


Fig. 12-13. The DNA replication process

Replication consists of three steps: unwinding, base pairing, and joining. Replication begins when an enzyme, *DNA helicase*, breaks the hydrogen bonds between the nucleotides. This divides the DNA into two single strands. As the helix unwinds, new nucleotides are added to the parent strands by the enzyme *DNA primase*. The *DNA polymerase* then continues to add more complementary base pairs such that A binds only to T, and C only to G. DNA polymerase does proofreading to avoid any mistakes. At the end of the replication process, the enzyme *DNA ligase* seals any breaks in the new DNA strand. Both DNA polymerase and DNA ligase also repair the DNA strands when damaged by harmful radiation or toxic chemicals.

In summary, replication is needed to ensure that all the body cells carry the same genetic material and that instructions are copied exactly for the next generation.

Transcription: Making Working Copies of the Genes

An architect protects the original blueprint of a building in a safe place by only giving copies of the blueprint to his on-site workers. Similarly, the DNA instructions are kept inside a safe place: the nucleus. Since the DNA does not leave the nucleus, it only sends copies of the blueprint of particular genes out of the nucleus to direct the assembly of a particular protein. It is the RNA's job to make blueprint copies of the DNA's instructions. Three kinds of RNA are involved in the process of protein synthesis, namely, the messenger RNA (mRNA), ribosomal RNA (rRNA), and transfer RNA (tRNA).

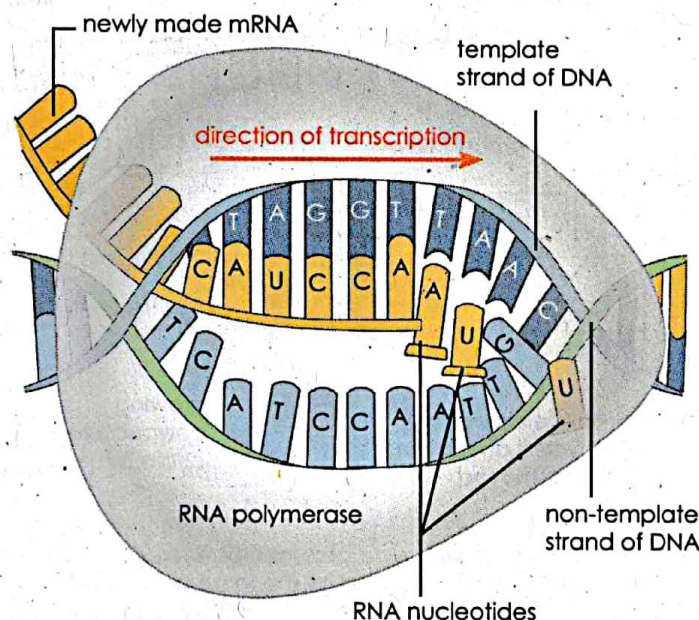
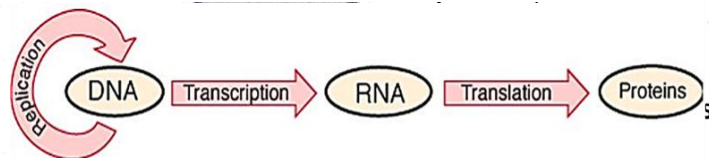


Fig. 12-14. Close-up view of how transcription produces genetic messages from the DNA in the form of RNA

In transcription, the mRNA copies specific instructions from the DNA in the nucleus so that it can bring the information to the cytoplasm for the next step. Transcription begins inside the nucleus when the DNA unzips between its base pairs. A portion of the DNA serves as a template for mRNA formation. The enzyme RNA polymerase initiates the DNA transcription, ensures that the right sequences are transcribed, and produces a complementary strand. To form the RNA strand, mRNA bases pair up with the existing DNA bases. This process is similar to replication in that cytosine pairs with guanine. But uracil, and not thymine, pairs up with adenine. The mRNA leaves the nucleus with the copy of the genetic instructions and enters the cytoplasm.



Central dogma of molecular biology

Translation: Making the Protein

Translation is the last stage in gene expression, which leads to the formation of protein. Specifically, translation is the process involved when genetic information is used to create amino acids and the corresponding proteins. Inside the cytoplasm, the mRNA attaches to a ribosome to provide the code for the specific protein that will be made. The three-base code in the mRNA is called a *codon*. During the process, the ribosomes move along the mRNA strand where the codons will be read and translated. The tRNA attaches to the mRNA inside the ribosome. Acting as an interpreter, the bases on the tRNA, called *anticodon*, read and translate the message by pairing up an equivalent three-letter code to the codons of the mRNA. As the codon is read, tRNA brings the appropriate amino acid to the ribosome. Each amino acid is represented by certain codons.

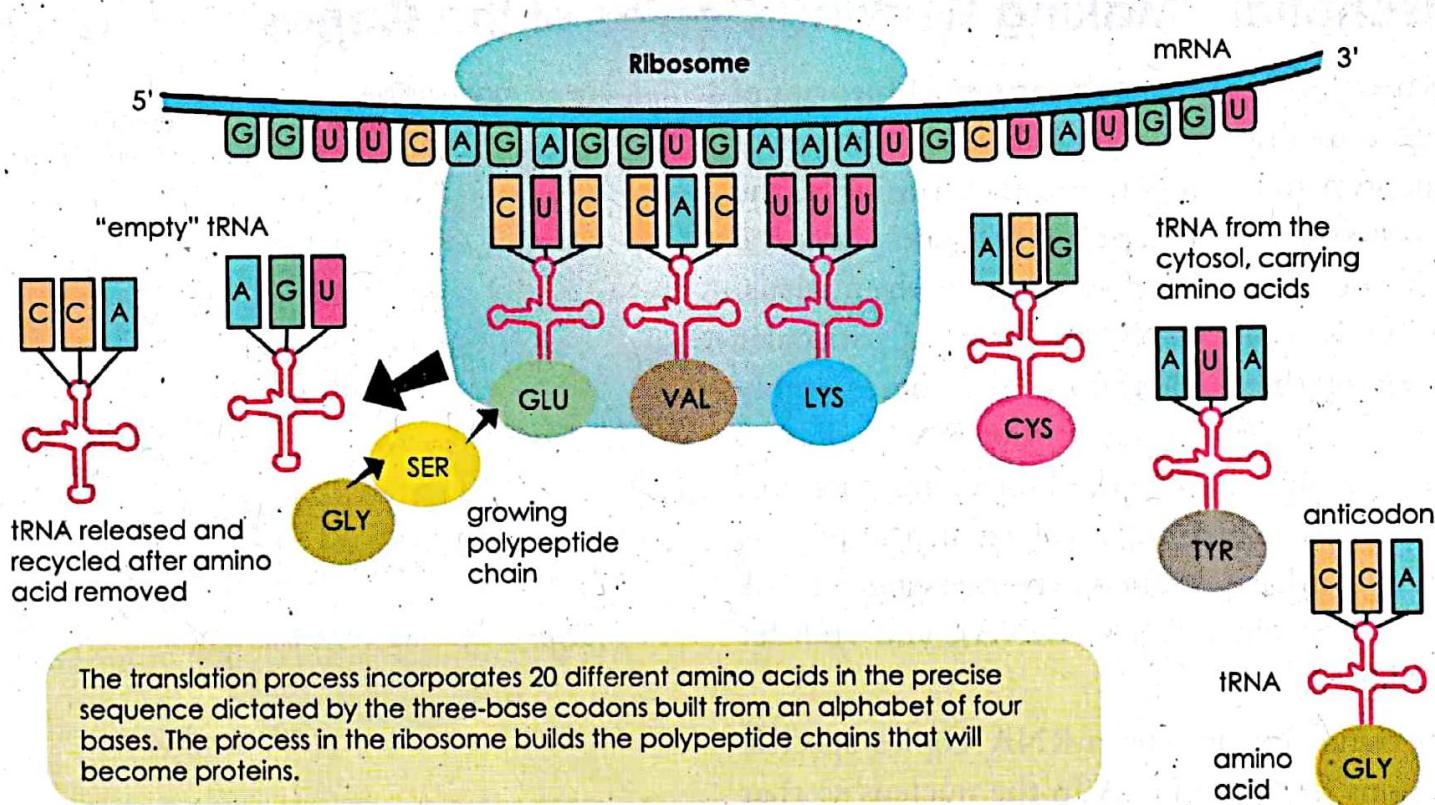


Fig. 12-15. How a message from the RNA is translated into proteins

The universal language that translates the gene and the amino acid is the genetic code, which matches the three-letter combination of the four nitrogenous bases to its equivalent amino acid. For example, the codon GUA (with an anticodon CAU) will code for the amino acid valine. A codon (mRNA) is complementary to the anticodon (tRNA).

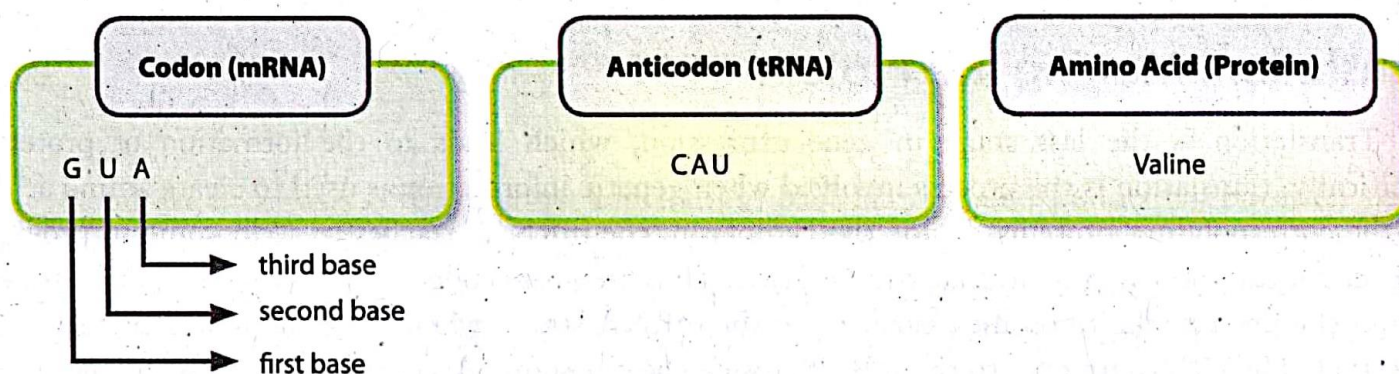


Fig. 12-16. Translation of codon and anticodon to amino acid

		Second Base				
		U	C	A	G	
First Base	U	UUU } Phenyl-alanine UUC } UUA } Leucine UUG }	UCU } UCC } Serine UCA } UCG }	UAU } Tyrosine UAC } UAA stop codon UAG stop codon	UGU } Cysteine UGC } UGA stop codon UGG Tryptophan	U C A G
	C	CUU } CUC } Leucine CUA } CUG }	CCU } CCC } Proline CCA } CCG }	CAU } Histidine CAC } CAA } Glutamine CAG }	CGU } CGC } Arginine CGA } CGG }	U C A G
	A	AUU } Isoleucine AUC } AUA } Methionine start codon AUG }	ACU } ACC } Threonine ACA } ACG }	AAU } Asparagine AAC } AAA lysine AAG }	AGU } Serine AGC } AGA } Arginine AGG }	U C A G
	G	GUU } GUC } Valine GUA } GUG }	GCU } GCC } Alanine GCA } GCG }	GAU } Aspartic acid GAC } GAA } Glutamic acid GAG }	GGU } GGC } Glycine GGA } GGG }	U C A G

Fig. 12-17. The genetic code is used to determine the amino acid specified by a codon. The first base of a codon is on the left side of the genetic chart. The second base is at the top, and the third base is on the right side of the chart.

The amino acids are then joined by peptide bonds to form proteins. The protein chain grows as more amino acids are attached.

In humans, only 20 amino acids are produced. Further processing will happen before the protein becomes functional. After these processes, the protein is used by the body for the various structural and physiological functions. These proteins will then interact with other proteins and molecules in the cells in order to create an observable characteristic called phenotype (e.g., eye color, skin color, hair type).

Lesson 13.1

Evolution: A Change Over Time

Main Idea: Evolution is an endless process that drives organisms to adapt to their environment.

EQ: Why do you adapt to your environment?

In the wild, the Satanic leaf-tailed gecko (*Uroplatus phantasticus*) is unmistakably unspotted by its predators since it almost looks like a dead leaf. The leaf-tailed gecko's unique characteristics and the rest of the animals that can do such amazing feat illustrate three observations by wildlife biologists:

1. organisms are well suited to the environment where they live;
2. although it appears to be different, it shares a lot of similar common characteristics with other life forms; and
3. it offers a glimpse of life's rich biodiversity.

A century and a half ago, a young naturalist also made several remarkable observations in his journey across continents. He was puzzled by his observations: Why are there many kinds of organisms in the planet of different shapes, sizes, behaviors, and habitats? How did they arise? Are they related?



Fig. 13-1. Satanic leaf-tailed gecko disguises as a leaf in order to catch prey and avoid predators

In 1858, Charles Darwin published *On the Origin of Species* containing compelling evidence on evolution. He would later come to be known as the greatest biologist of all time. He ushered a scientific revolution known today as the era of evolutionary biology.

In the past, "evolution" was not a common term. Darwin coined the phrase *descent with modification* to propose that present-day organisms are descendants of ancestral species. Today, *evolution* is used to refer to any change in the genetic composition of a population of organisms over time. The evolutionary process takes a long period of time—some requiring millions of years. Recent studies have shown that some organisms such as bacteria evolve rapidly. This is because of their adaptive characteristics that allow them to withstand the fatal effects of drugs, multiply in minutes, reignite infections, and infect more hosts. Antibiotic resistance is a startling example of evolution.

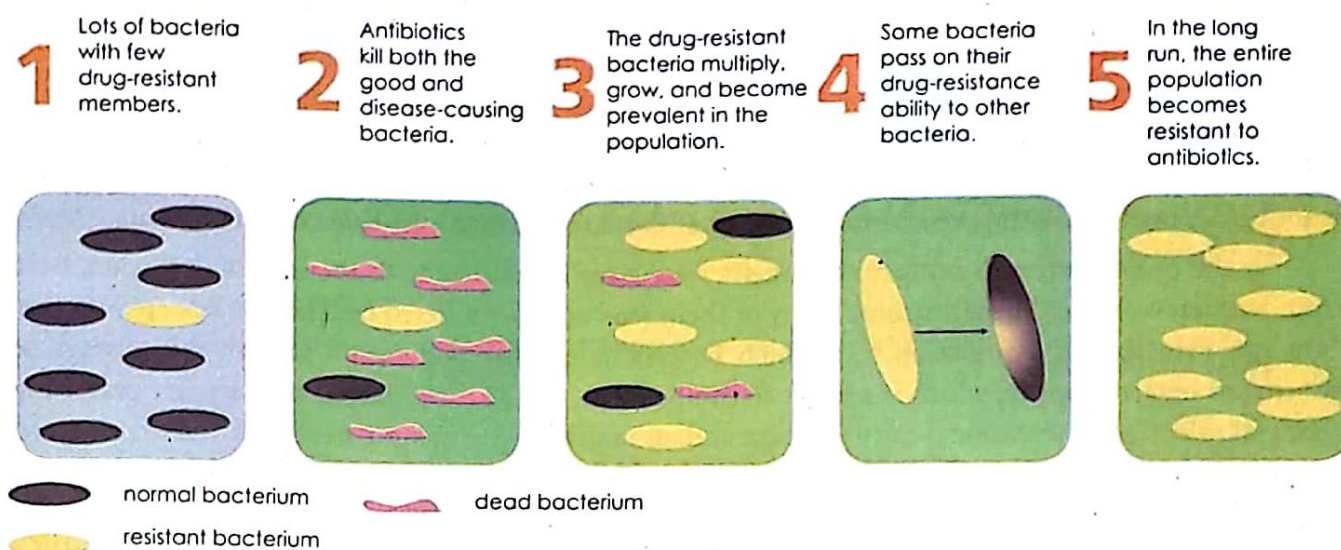


Fig. 13-2. How bacteria become antibiotic-resistant

Biological evolution is a powerful and important process that lasts for billions of years. It gradually selects the organisms that are better adapted to their environment. Evolution will continue for as long as organisms are born, die, and compete for resources they need to survive and reproduce.

Darwin's Theory of Natural Selection

Charles Robert Darwin, an English naturalist and geologist, is credited for developing the *theory of natural selection*, which basically posits the "survival of the fittest." In fact, before he even developed his own ideas about evolution, other scientists had already proposed that organisms change over time.

Interestingly, one of them was his own grandfather, Erasmus Darwin, and another scientist, Georges-Louis Leclerc (Comte de Buffon), but both could not explain how or why organisms evolved. Both kept their ideas to themselves since they knew that such notions would not be accepted.

On December 27, 1831, Charles Darwin left England aboard a ship, *HMS Beagle*, to conduct scientific exploration. His five-year journey helped him gather evidence and formulate his idea about evolution as he became more and more convinced of natural selection. However, Darwin was also reluctant to publish his ideas because they would oppose to the prevailing ideas of that time.

Darwin's curiosity and analytical skills influenced his interest in natural history. He started to collect evidence in order to provide a scientific explanation for the diversity of life on the planet. His observations included patterns of diversity such as why certain organisms such as the ostriches of Africa can only be found in one place, but similar flightless birds can be found in South America and Australia.

Darwin also collected and studied fossils leading him to ask why these species disappeared and how are they related to present-day organisms. Darwin also became fascinated with the unique plants and animals he saw at Galapagos Islands, given that the islands were close to one another and yet have different climates and amount of vegetation. He observed that the tortoises in the islands slightly differ depending on the island the tortoise inhabited as shown in figure 13-7.

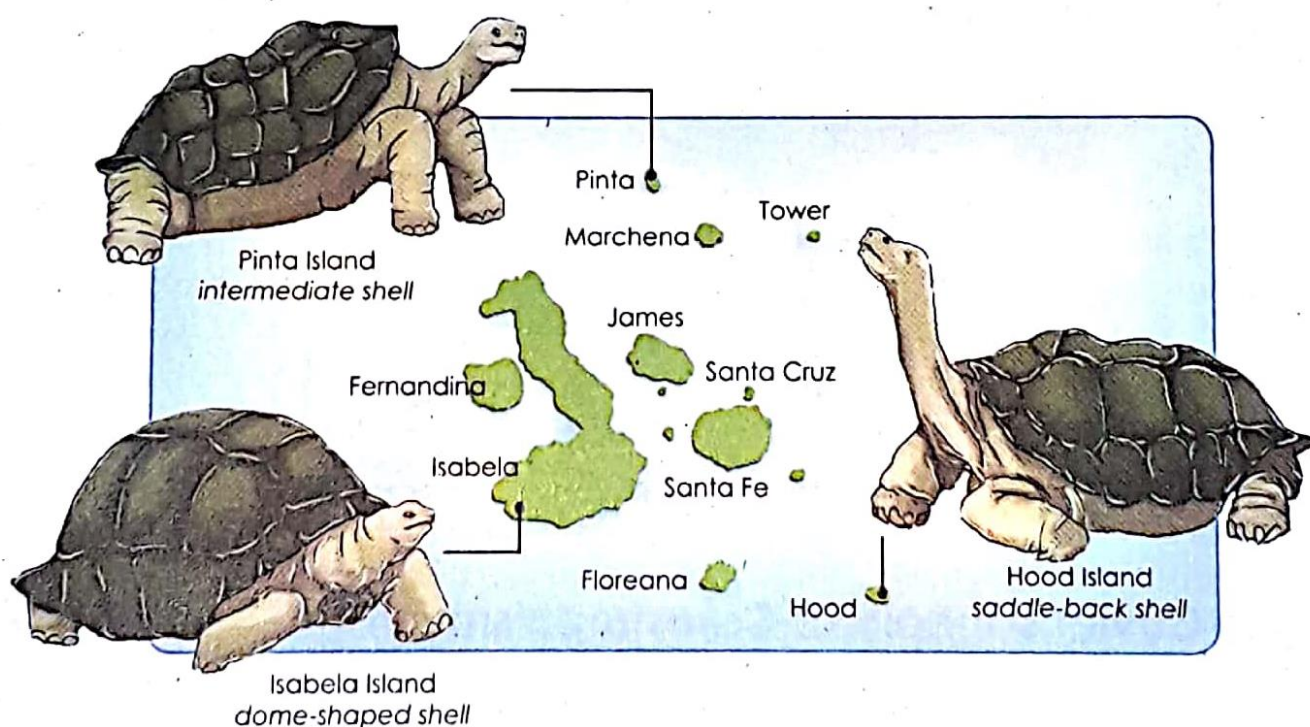


Fig. 13-7. Darwin observed that species characteristics vary between the Galapagos Islands, such as the different shapes of tortoise shells shown above, which correspond to different habitats. The Hood Island tortoise has a long neck allowing it to reach to sparse vegetation, while the Isabela Island tortoise has a dome-shaped shell and shorter neck because the vegetation in the island is closer to the ground. The Pinta Island tortoise has a shell in intermediate of the two other tortoises.

In 1858, Darwin was prompted to publish his work after reading "On the Tendency of Varieties to Depart Indefinitely from the Original Type," an essay written by British naturalist Alfred Russel Wallace (1823–1913), who had come up with a similar theory. Both Darwin and Wallace knew nothing of genes and chromosomes, but they provided an explanation for both the diversity and similarities between living organisms.

On November 24, 1859, Darwin published his book, *On the Origin of Species by Means of Natural Selection*. It outlined his theory of evolution, which was greatly influenced by the biological, geological, and geographical observations he made during his five-year journey. Darwin's ideas were greeted with enthusiasm but he still had to defend his ideas from scientists for lack of hard evidence and outcry from people comparing evolution to heresy. Despite the controversies, his theory slowly gained acceptance from biologists around the world. Today, new insights in the field of genetics have improved understanding of how natural selection brings about evolution. Evolution is now a well-accepted basis for the diversity of life on Earth. The *Origin of Species* fueled the expansion of biological research and knowledge that continues today. In this regard, evolution unifies many fields within biology.

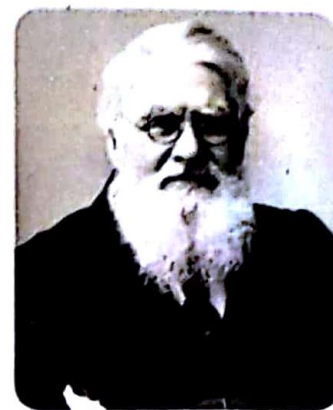


Fig. 13-8. Alfred Russel Wallace studied the flora and fauna of South America, Malay, and the East Indies. Wallace published his own thesis on evolution by natural selection.

What Darwin Believed

Darwin believed that the needs of animals have nothing to do with how they evolve and that changes in an organism during its lifetime do not affect the evolution of the species. He said that organisms, even of the same species, are all different and that varying traits help them survive and have more offspring. The offspring are born with their parents' helpful traits, and as they reproduce, such adaptive characteristics prevail in more of the species. Other individuals that have not adapted will die off.

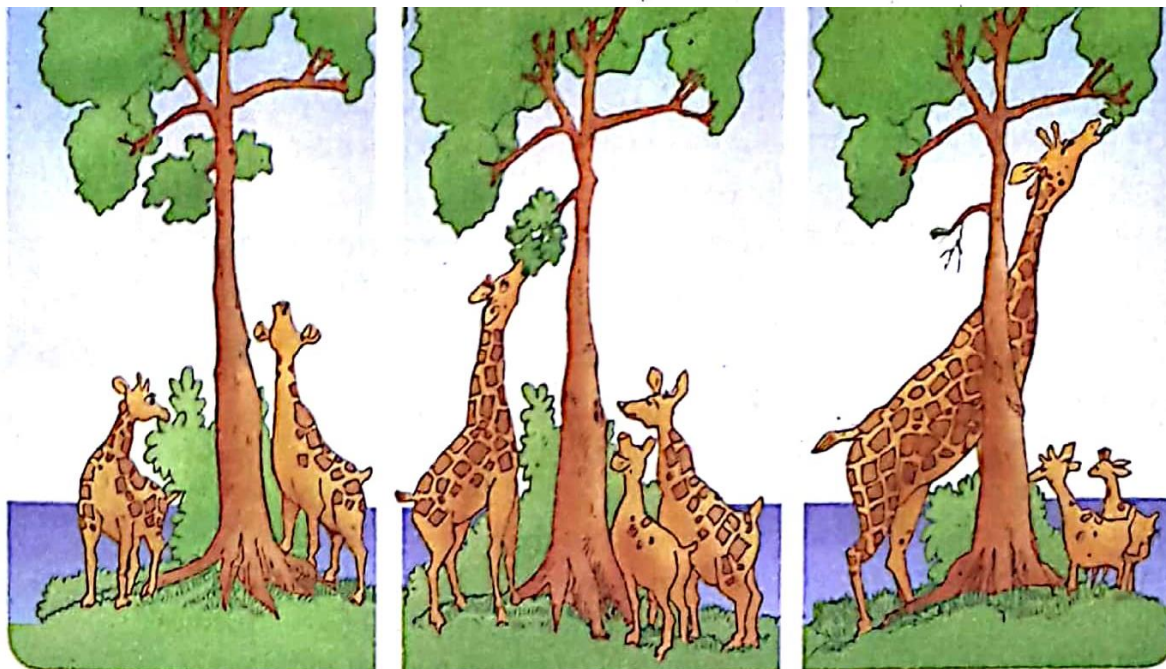


Fig. 13-9. A representation of a giraffe's evolution according to Darwin

According to Darwin, giraffe ancestors had shorter necks. Some giraffes with chance mutations or with different traits produced longer necks. This adaptive trait enabled long-necked members of the populations to avoid competition with other animals, including other giraffes with short necks. Having long necks enabled some giraffes to obtain plenty of food to survive and to reproduce. Natural selection enhanced the survival of long-necked giraffes. Competition for food drove short-necked giraffes to extinction. Ultimately, only giraffes with long necks remained.

Lesson 13.3**Mechanisms
of Natural Selection****Main Idea:** Natural selection explains evolutionary changes among organisms.**EQ:** How does natural selection lead to evolution?

During his voyage around the world studying plants and animals, Charles Darwin observed that reproduction roughly balances death with illness, accident, and old age. He noted the different characteristics even within members of the same species, and observed that many of these variations were inherited. He proposed that individuals whose characteristics allow them to survive and to reproduce would leave behind more offspring than less well-adapted members. Consequently, these characteristics are more widespread in the subsequent generations. This illustrates the idea of natural selection. Darwin proposed that over many generations, the characteristics of a population could change so significantly that a new species altogether is formed. Natural selection attempted to explain the causes of evolutionary change. The theory has been divided into five parts, each of which explains how evolution happens.

Overpopulation

The number of offspring is usually greater than the available resources necessary for organisms to survive. Organisms can die from many causes—disease, starvation, or being eaten by predators. The environment cannot support every organism. Many die before they can reproduce.

Loggerhead sea turtles, for example, may have three to eight nests every season, with each nest containing about 100 Ping-Pong ball-like eggs. Only about one out of a thousand lives on to reproduce because high tides, beach erosion, and plant roots can destroy a nest and thus kill the others. Predators such as raccoons, crabs, foxes, and seabirds also eat the eggs. Upon hatching, the young turtles face these predators again. Some may even get stuck on the beach and die. If all of the eggs hatched and each of them laid as many eggs, there would be too many turtles for the environment to support.



Fig. 13-10. Turtle hatchlings taking their first steps into the ocean

Struggle for Existence

Every organism must struggle to survive. One reason that organisms survive is that they compete for limited resources, including food, territory, and other necessities of life. They also struggle to elude predators and to overcome diseases. Competition may be active, as in the battle between two predatory animals over a single prey; or passive, as in two wild flowers in a bed of grass competing for sunlight, water, and space.

The mass migration of 1.5 million wildebeests, together with half a million zebras and gazelles on their perennial search for water and greener pasture, is one of the greatest natural spectacles on Earth. From June to July after the rainy season, these animals head west to cross the Serengeti River in Kenya.



Fig. 13-11. A Nile crocodile attacks a wildebeest and prepares to drag it underwater.

The weak, crippled, or young wildebeests become easy targets for the numerous, flesh-hungry crocodiles that notoriously roam the river. Both crocodiles and wildebeests struggle to survive. The crocodiles live on but the wildebeests do not. However, the fitter wildebeests continue the journey and the propagation of their species.

Polar bears have a thick layer of wooly fur that is different from that of any other animal. This fur resembles plastic straws that keeps the bear's hair from matting down while it swims in the cold arctic water. The white appearance of the polar bear's fur helps camouflage it as it hunts for its prey, such as seals and walruses. Underneath the fur is black skin that helps it absorb heat from the sun and stay warm.



Fig. 13-12. This polar bear has thick, wooly fur to keep it warm and help it survive frozen polar regions.

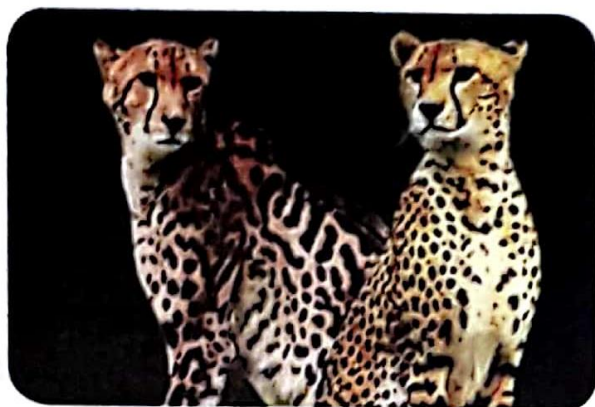


Fig. 13-13. Variations in spots of cheetahs



Fig. 13-14. Variations in skills among horses

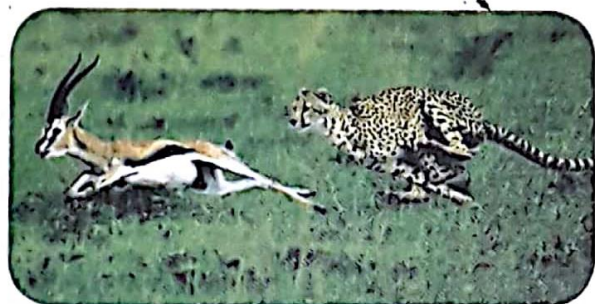


Fig. 13-15. A race for survival between a cheetah and a gazelle

Presence of Variation

There is variation within a species. Traditionally, a *species* refers to a group of organisms that share similar characteristics and interbreed to produce fertile offspring. There are physical variations or differences within members of a species. Cheetahs in a pride would differ from one another. If you look at their spots, for instance, you will notice that they do not have the same color streaks or arrangement of spots.

Aside from physical differences, differences in skill and behavior may arise, such as the speed of horses. If organisms were all the same, none would be better suited than any other, and selection cannot occur. Those that survive the competition possess variations that give them an advantage.

Survival of the Fittest

Some variations allow members of a species to survive and reproduce better than others. Certain traits help organisms survive and reproduce in a given environment. Naturally, a gazelle is at a disadvantage against a cheetah, the world's fastest land mammal, which can run at 96 kilometers an hour. However, gazelles can outlast cheetahs and can sustain their speeds for long distances. A quick cheetah is more likely to catch a gazelle, but a gazelle is more likely to escape the cheetah if the encounter drags on over long distance.



Fig. 13-16. A caterpillar full of bristles with venom glands is more likely to be left alone by predators.

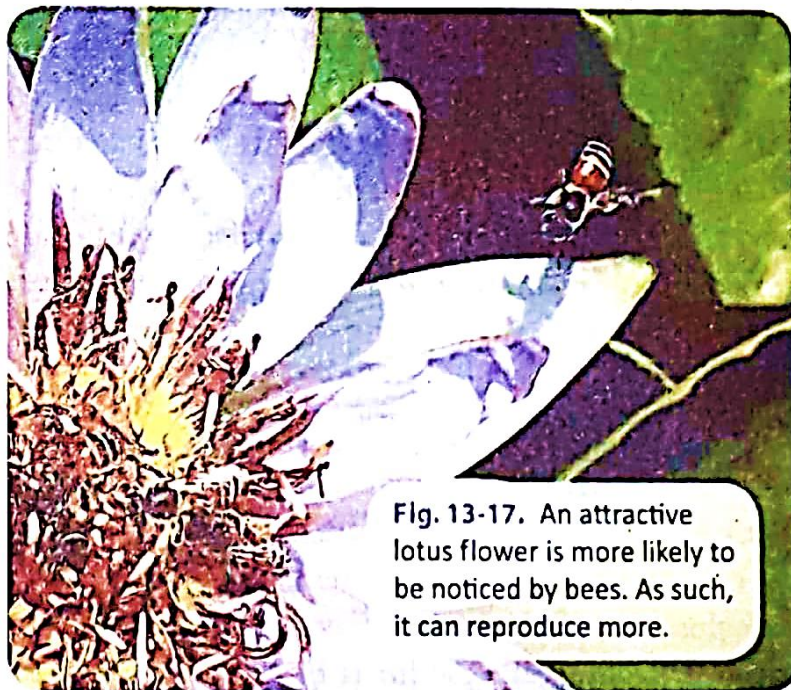


Fig. 13-17. An attractive lotus flower is more likely to be noticed by bees. As such, it can reproduce more.

Variations Can Be Transmitted

Organisms that survive and reproduce pass on their traits to their offspring, and the helpful traits gradually appear in more and more of the population. Organisms with traits that are well-suited to the environment survive, reproduce, and successfully pass on their traits to their offspring. These traits enable organisms to "fit" into their environment, while those who lack these variations slowly die off. If more individuals of the population has the advantageous trait, it will be passed on to the succeeding generations at an exponential rate.



Fig. 13-18. The faster cheetah can catch its prey and survive.

Faster cheetahs will be able to catch food more often and survive; the slower ones die out. As more faster cheetahs reproduce over generations, their population will gradually be only made up of faster members.

ADDITIONAL NOTES:

Lesson 14.1**Flow of Energy
and Matter in Ecosystem**

Main Idea: Organisms interact with their environment in a process involving the exchange of matter and energy.

EQ: How dependent are you on other organisms and the environment?

Recall how organisms interact with the physical components in the environment, with other species, and with their own kind. Figure 14-1 shows the relationship of the processes that power life on Earth—the exchange of matter and of energy. Plants acting as producers take water and minerals from the soil and carbon dioxide from air. During photosynthesis, plants capture solar energy to convert carbon dioxide and water into sugar and oxygen. The consumers, on the other hand, eat plants and other animals. To release energy from food through cellular respiration, they must take in oxygen and release carbon dioxide, while animal waste return to the environment. Small animals, fungi, and bacteria act on and decompose wastes and the remains of dead organisms. The decomposers recycle these materials into simpler forms of nutrients that plants can use. Thus, matter is cycled throughout the environment.

Energy in an ecosystem flows in one direction, starting with plants that absorb and convert light energy into chemical energy (sugar). When animals eat other organisms, the chemical energy is passed on to the consumers. During energy conversion, some energy is transformed into heat. Therefore, the exchange of matter and energy in an ecosystem requires constant interactions, from the sun's one-way flow to the cycling of nutrients between producers, consumers, and decomposers.

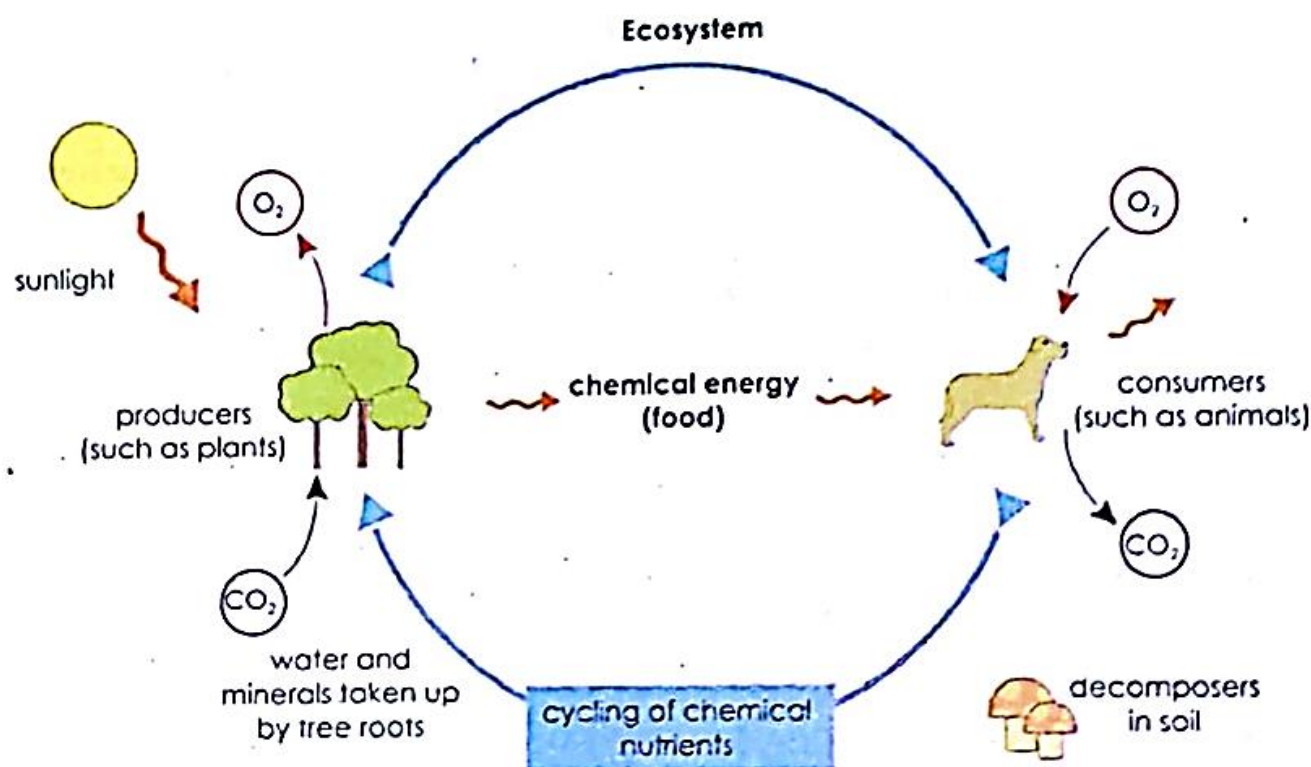


Fig. 14-1. The flow of energy and nutrient cycling are processes vital to the survival of populations.

Lesson 14.2

Population Dynamics

Main Idea: The capacity for growth is a measure of the prospects for survival of a population in a particular habitat.

EQ: How do you describe the population dynamics of your community?

You have probably heard of stories about the janitor fish (*Pterygoplichthys pardalis*) and its burrowing habit, infesting and destroying the Marikina Riverbanks for years. Ever since, how it was introduced in the Philippines has been the subject of much speculation. Its relative, *P. disjunctivus* has gone far to invade Laguna de Bay and the Agusan Marsh in Mindanao. Many introduced fish species have reached our waters since 1905 for the purpose of increasing food production, recreational, and ornamental purposes without the authority of Bureau of Fisheries and Aquatic Resources (BFAR) under the Department of Agriculture. Many of these species, whether accidentally or deliberately introduced, have become invasive and caused economic losses and ecological damage to the native environment. Meanwhile, the fishermen who make a living by catching staple fish such as *tilapia* along the river face the repercussion of janitor fish invasion. Despite their efforts of removing these invasive species, their catch continues to drop dramatically. Can anything be done to increase the number of staple fish? How can the janitor fish be completely eradicated in local waters? To better understand why populations change such as the janitor fish population growing incessantly or why the *tilapia* population is plummeting down, one should study population ecology.



Fig. 14-2. A poster being disseminated by the Bureau of Fisheries and Aquatic Resources (BFAR) to educate the public on invasive species that threaten Philippine lakes and fisheries.

The study of life can extend from the global scale of the biosphere to the microscopic component of an organism. Different ecosystems house different sets of organisms that make up a *community*. A group of organisms having the same characteristics belongs to a *population*. Population ecologists study the unique characteristics of populations that distinguish them from one another. The different studies about populations of organisms belong to an umbrella of knowledge known as *population ecology*. It is important for ecologists to know the stability of an ecosystem based on populations. This knowledge can be a guide toward a better, more efficient management of resources, which, in turn, helps to avoid problems such as species extinction, population explosion, lack of food supply, or pest infestation.

END OF TOPIC

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February 05, 2020